Reward Shaping Is Noise Shaping, and a Law for the Accuracy–Diversity Frontier of RLVR

Project report: what is established, what is open, what is running

Mohamed Ayman

October 23, 2026

The project in five points

1. **Question.** RLVR methods that change how answers are scored claim more diversity (pass@k), but they are compared at the same training step. Is the diversity real?
2. **Diagnosis (established).** Every correctness-only scoring rule pushes the model in the same expected direction and differs only in per-question weight and noise; noise sets the *speed* along one accuracy–diversity curve. At matched accuracy, the rules coincide on two model families, and zero-mean noise alone reproduces NSR's behaviour.
3. **Methods (so far small).** Replaying successes, inside RL, gives about +1 point of pass@16 at matched accuracy. A warm start on base-model successes and groups of 32 rollouts go far beyond Dr. GRPO's accuracy range, so they cannot yet be compared at matched accuracy.
4. **New: a frontier law.** Starting only from the base model's per-question success counts, a two-parameter model of per-question log-odds learning predicts pass@16 within 1–2 points for nine further training rules, a second model family, and a larger answer budget (pre-registered, prospective). It fails after a warm start.
5. **Status.** Independent review (Codex): the diagnosis alone about 5/10; with the law passing its prospective tests about 7/10; with a method that improves the frontier across scales about 8/10. The long-run test of the law, the 2026-methods audit, and the method pilot are running.

1 Setting and terms

Qwen2.5-Math-1.5B (main) and OLMo-2-1B (replication), MATH training questions, 150 RL steps of 16 questions $\times$ 8 answers, AdamW 10^{-6}, answers up to 512 tokens; evaluation on 511 held-out MATH test problems (answer keys audited) with 32 samples per problem.

pass@1 / pass@k	accuracy / chance that at least one of k answers is right ("diversity")
Dr. GRPO (B)	advantage $A = r - \bar{r}$; the reference method
PSR / NSR	reward right answers only ($A = r$) / penalize wrong answers only ($A = r - 1$)
Matched accuracy	pass@16 compared at the same pass@1, interpolated along B's trajectory (and its temperature envelope)

2 The diagnosis (finished)

Theory. For any rule that scores answers by correctness alone, with p a question's success rate, $\mathbb{E}g = c(p)\,\nabla p$ and $\mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p)$: a rule can only reweight questions (c) and change the noise (D_1, D_0). Wrong answers carry most of it (2–3$\times$ larger squared gradients, measured on both families). At matched pass@1, pass@$k = \mathbb{E}_x[1 - (1 - p_x)^k]$, so diversity at equal accuracy is how evenly success is spread across questions.

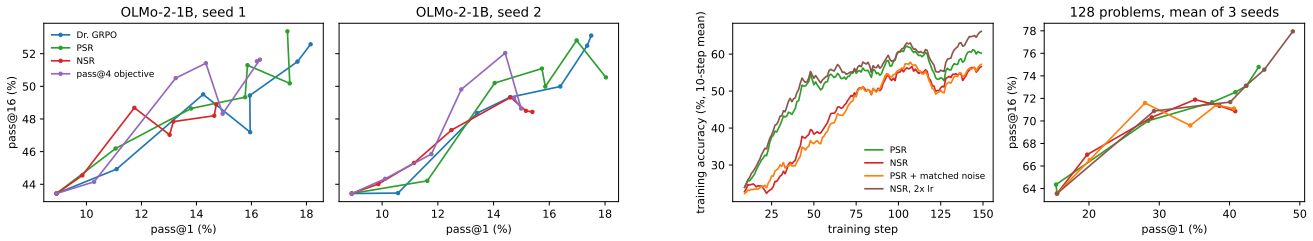


Figure 1: Left: OLMo-2-1B, five rules on one accuracy–diversity curve. Right: Qwen; PSR plus zero-mean noise of NSR's size learns as slowly as NSR, and all lie on one curve.

Established findings

1. **One curve.** PSR, NSR, five constant baselines, the NSR paper's own implementation and a pass@4 objective all move along one curve on Qwen; on OLMo, NSR minus PSR at matched accuracy is -0.30, 90% $[-1.28, +0.77]$ (equivalent within ± 2).
2. **Noise is the cause.** Adding zero-mean random-sign noise of NSR's size to PSR makes it as slow as NSR (speed $\phi = -0.11$ $[-0.22, -0.05]$; NSR $= 0$, PSR $= 1$); NSR at twice the learning rate recovers. With Adam, noise misdirects rather than shrinks steps.
3. **Not a temperature effect.** RL barely changes entropy on fixed text ($1.081 \to 1.064$) but cuts it 3–4× on its own samples; a late checkpoint sampled hotter does not beat the training curve (-1.42, 90% $[-2.19, -0.68]$).
4. **Popular fixes stay on the curve:** entropy bonus, adaptive entropy and entropy floors (which stop learning), and the published unlikeliness reward ($+0.14$).

3 Trying to move the curve

Method (3 seeds, 511 problems unless noted)	pass@16 at matched accuracy	Status
Reward shaping, entropy control, unlikeliness, pass@k objective	≈ 0	finished
Coverage-weighted replay of own successes (Pre-reg. 12)	$+1.19$ $[+0.54, +1.79]$	finished; below +2
Base-seeded replay, Option A (Pre-reg. 14)	$+1.02$ $[+0.23, +1.68]$; pass@1 -5.4 at step 150	finished (main arms)
Annealed replay; MaxRL with group completion (pilot, 1 seed, validation)	$+1.30$; $+1.12$; both pass@1 -6.5	pilot, interim
Group completion; prefix-guided rollouts (pilot)	–	being evaluated
Warm start: fine-tune on base successes, then Dr. GRPO	beyond B's range (pass@1 52.5 vs 44.1)	not comparable yet
Groups of 32 rollouts (4× samples)	beyond B's range (pass@1 52.5)	2 of 3 seeds

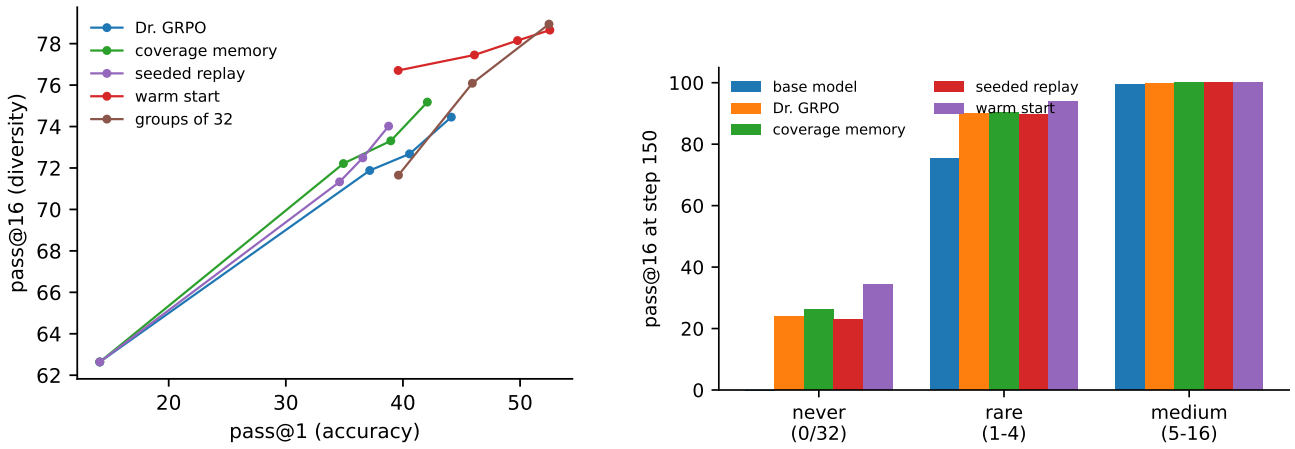


Figure 2: Left: the Qwen frontier (seed means). The warm start and groups of 32 end at the same point (pass@1 52.5, pass@16 about 79), beyond anything Dr. GRPO reaches in 150 steps. Right: pass@16 at step 150 by how often the base model solved each problem; the warm start's gain sits on problems the base never solves (24 → 34).

The 512-token budget matters. At 512 tokens, 48% of Dr. GRPO's sampled answers at step 150 are truncated. Re-evaluated at 1,024 tokens (Check 16, partial), Dr. GRPO's step-150 pass@1 is 57.7 (vs 44.6) and pass@16 88.4 (vs 75.2), with 15.5% truncated; the warm start reaches 63.8 and 89.7 (4.9% truncated). Every absolute number in the project depends on the budget; the comparisons are made within one budget, and the law (Section 4) was checked at both.

4 The frontier law (Pre-registration 17)

The law

Each question's log-odds start at the base model's empirical-Bayes estimate $\theta_x(0) = \log \frac{c_x+a}{n-c_x+b}$ and grow as $\dot{\theta}_x = p_x(1-p_x) + \tau\,\overline{p(1-p)}$: own-signal logistic learning plus a uniform transfer term. A checkpoint is placed where the predicted mean success equals its observed pass@1; the law then predicts the rest of the distribution (pass@k, difficulty strata, per-question counts). Two parameters: progress t (matched) and τ (fitted once on Dr. GRPO, $\tau = -0.20$, then frozen).

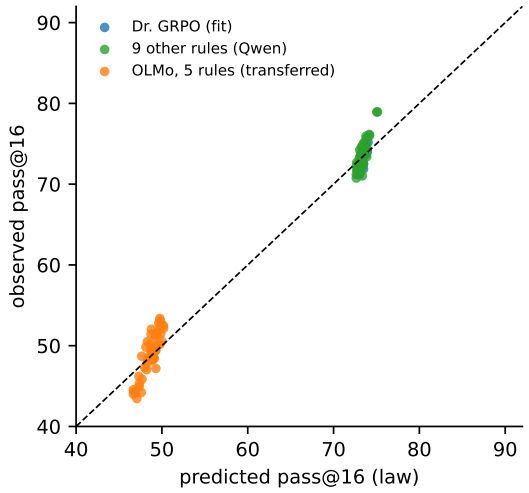


Figure 3: Predicted against observed pass@16. Blue: Dr. GRPO (fit). Green: nine further rules on Qwen. Orange: OLMo, five rules, τ transferred from Qwen without refitting.

Test set (law frozen before each)	ckpts	RMSE: law / unif. / affine	verdict	beats affine
Fit: Dr. GRPO (in-sample)	9	1.06 / 21.3 / 1.28	–	9/9
T1: nine other rules on Qwen	56	**1.21** / 20.1 / 2.45	supported	54/56
T4: OLMo, τ from Qwen	50	**1.90** / 9.9 / 12.6	supported	50/50
T5: 1,024-token budget	4	**0.85** / 11.5 / 1.27	supported	4/4
T2: warm start, from its own step-0 counts	9	3.56 / 9.0 / 8.0	falsified	9/9
T3: 450-step Dr. GRPO (beyond the fitted range)	–	running	–	–

"Uniform": every question's log-odds shift by the same amount; "affine": $\theta' = \alpha + \beta\theta$ with β fitted on the same data. "Beats affine": higher per-question likelihood of the observed counts.

Reading. From a base model, the frontier of every rule we ran, on two families and two budgets, is predicted by the base difficulty distribution plus near-independent per-question learning; rules differ in how far along the trajectory they get. Diversity collapse is a rich-get-richer effect: a uniform improvement would lift never-solved problems to about 70–80 pass@16, the observed value is 24. Two levers can move this frontier: the starting distribution (information about rare questions) and transfer between questions. **The exception**: after a warm start, RL lifts never-solved problems about 12 points less than the law predicts (refitted $\tau = -0.65$ to -0.70 in all seeds, against -0.20 from the base model): fine-tuning changes the dynamics, not only the starting point.

Against the law's extended frontier (the law from base counts, used beyond Dr. GRPO's range), the warm start is +3.3 to +3.6 points above at every checkpoint and groups of 32 reaches +3.9 at step 150; Dr. GRPO itself drifts from -1.2 to $+0.5$. Whether the warm start is genuinely above Dr. GRPO's curve depends on how the law extrapolates, which the 450-step run (T3) measures.

5 Does training transfer to rarely solved questions? (Check 17, partial)

For 395 held-out problems, the margin between the likelihood of the base model's correct and wrong answers (style cancels). Seed 8 of Dr. GRPO: the margin rises by +0.025 nats per token on both rarely solved and medium problems by step 150; the warm start (seed 9) is +0.039 above the base on rare problems right after fine-tuning and +0.049 by step 150. Interim reading: Dr. GRPO does transfer to rare problems about as much as to medium ones, so the "little transfer" premise is not supported in this seed; four of six lanes are still running.

6 Position in the 2026 literature

- **Unpublished (ours):** noise sets speed along one curve; NSR on PSR's curve on two families; the noise-injection causal test; the matched-accuracy protocol; temperature does not beat the curve; a prospective law for the frontier from base counts. No 2026 method paper compares at matched accuracy.
- **Published:** "same direction, per-question weight" (Davis & Recht 2025; Heckel et al. 2026); base-seeded replay (TAR, Oct. 2026); nearly our coverage weight (BBG); replay and off-policy rescue of all-wrong groups (LSPO, LUFFY, ExGRPO); transfer-driven curricula (TAC, GradAlign); per-prompt GRPO recurrences (Mroueh 2025); the Matthew effect (NGU); "RLVR is a Kernel", which argues base success does not determine post-RL success and is the strongest challenge to the law.

7 Assessment and next steps

Open questions and what decides them

1. **Does the law extrapolate?** T3 (450-step Dr. GRPO, seed 13 on AWS; 14 and 15 on Tamia) and the deep base sampling of the 309 hardest problems (which identifies the sign of τ).
2. **Is the warm start (and groups of 32) above the frontier, or only faster along it?** Decided by T3's extended reference and by Check 16 at 1,024 tokens.
3. **Which method?** The pilot's group completion and prefix-guided rollouts (being evaluated); the law suggests a method that allocates demonstration acquisition where it predicts stalls, compared with random allocation and the warm start at equal compute (needs its own novelty check).
4. **The 2026 audit:** TAR (3 seeds trained, being evaluated), BBG, MaxRL, W-REINFORCE (partly trained) at matched accuracy, and against the law.

Paper shape if T3 passes: the diagnosis (one curve, noise sets speed, matched accuracy) → the law (the frontier predicted from base difficulty; why reward shaping cannot move it; the warm-start exception) → the matched-accuracy audit of 2026 methods → a method derived from the law. Independent review estimates 7/10 for this shape if the prospective tests pass.

A Experiment ledger

	Question	Outcome
P1–3	Scale down wrong answers; collection rules at equal cost; noise inside training	noise formula fits; the optimal collection rule moves faster along the curve
P4	PSR vs NSR, clean on-policy	NSR 6.4× noisier, slower, same pass@64: separation not reproduced
P5	NSR vs PSR at matched accuracy	no extra diversity (-2.1, -0.2)
P6	The NSR paper's implementation	not recreated; clipping active on 26–28% of NSR's steps
P7	Same mean, graded noise (baselines)	speed follows predicted noise (rank correlation 0.9–0.97)
P8	pass@4 objective; reasoning graded	curve fixed ($+1.6$, $+0.2$); reasoning-checked diversity equal
P9	OLMo-2-1B replication	one curve: NSR−PSR -0.30 $[-1.28, +0.77]$
P10	Is noise the cause?	yes: PSR + noise as slow as NSR
P11	Is the curve a temperature effect?	no (-1.42 $[-2.19, -0.68]$)
P12	Coverage-weighted replay	$+1.19$ $[+0.54, +1.79]$, average agreement
P14	Base-seeded replay and controls	$+1.02$; slower; warm start far ahead
P15	2026 methods at matched accuracy (TAR, BBG, MaxRL, W-REINFORCE)	running
P16	Push mechanisms inside RL (pilot)	interim: replay variants ≈ $+1.2$ but 6.5 points less accurate
P17	The frontier law	supported on T1, T4, T5; falsified on T2; T3 running
Check 16	512 vs 1,024 tokens	all numbers rise at 1,024; comparisons pending
Check 17	Transfer to rare problems	partial: Dr. GRPO transfers to rare as to medium

All experiments were pre-registered and reviewed by an independent model before launch; the records are the pre-registration, results and check files in `experiments/training/`.